FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VII
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	24.	**B**	47.	**B**
2.	**C**	25.	**D**	48.	**B**
3.	**A**	26.	**D**	49.	**A**
4.	**A, C**	27.	**A, C**	50.	**B, D**
5.	**A, D**	28.	**A, B, C, D**	51.	**A, C**
6.	**A, B**	29.	**B, C**	52.	**A, D**
7.	**A, C**	30.	**A, C**	53.	**A, B**
8.	**A, B, C, D**	31.	**B, C**	54.	**B, C, D**
9.	**B**	32.	**C**	55.	**C**
10.	**C**	33.	**B**	56.	**A**
11.	**9**	34.	**3**	57.	**4**
12.	**6**	35.	**3**	58.	**8**
13.	**2**	36.	**4**	59.	**2**
14.	**6**	37.	**2**	60.	**5**
15.	**8**	38.	**6**	61.	**6**
16.	**4**	39.	**3**	62.	**3**
17.	**4**	40.	**2**	63.	**5**
18.	**8**	41.	**6**	64.	**1**
19.	**5**	42.	**4**	65.	**2**
20.	**5**	43.	**4**	66.	**4**
21.	**01800.00**	44.	**00004.00**	67.	**18564.00**
22.	**00720.00**	45.	**00004.50**	68.	**00038.00**
23.	**00004.00**	46.	**00001.25**	69.	**00501.00**

Physics

PART – I

SECTION – A

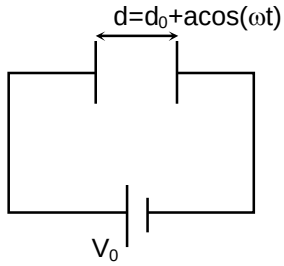
1. D

Induced charge $= q\left(1-\frac{1}{\epsilon_r}\right)$

Surface density $= \frac{q\left(1-\frac{1}{\epsilon_r}\right)}{4\pi r_2^2} = \frac{q(\epsilon_r - 1)}{4\pi\,\epsilon_r\, r_2^2}$

2. C

$I = \frac{d}{dt}Q = \frac{d}{dt}(CV) = V\frac{d}{dt}\left(\frac{\epsilon_0\, A}{d_0 + a\cos\omega t}\right)$



$I = \frac{V\,\epsilon_0\, A[a\omega\sin\omega t]}{(d_0 + a\cos\omega t)^2}$

$= \frac{V\,\epsilon_0\, A[a\omega\sin\omega t]}{[d_0 + a\cos\omega t]^2}$

When $\sin\omega t = 1, \cos\omega t = 0$, 'I' becomes maximum

$\therefore\ I_0 = \frac{V\,\epsilon_0\, A\omega a}{d_0^2}$

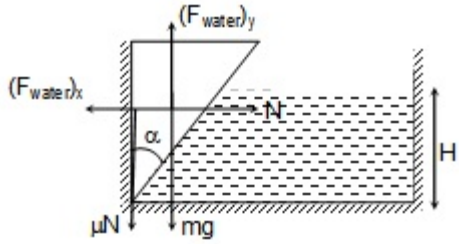
$a = \frac{I_0 {d_0}^2}{V\,\epsilon_0\, A\omega}$

3. A

$(F_{water})_y = \frac{H}{2}(H\tan\alpha)b\rho g$

$(F_{water})_x = \frac{H}{2}(\rho g)(Hb)$

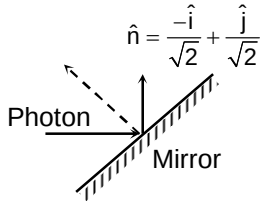
When 'H' is max friction will be maximum.
Use condition of equilibrium.



4. A, C

Change in momentum of each photon $= \frac{h}{\lambda}\left(-\hat{i} + \hat{j}\right)$

Force exerted by light beam on mirror $= \frac{P}{C}\left(\hat{i} - \hat{j}\right)$



5. A, D

Ring is at (–)ve potential and V(∞) = 0, electric field due to this potential difference will accelerate electron away from ring for maximum speed of photoelectron immediately after emission

$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}mV_{max}^2 \qquad \Rightarrow \qquad V_{max} = \sqrt{\frac{2hc}{m}\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)}$

For maximum speed of photoelectron at '∞'

$$\frac{hc}{\lambda}+eV_0=\frac{hc}{\lambda_0}+\frac{1}{2}mV'^2_{max}$$

$$V'_{max}=\sqrt{\frac{2hc}{m}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)+\frac{2eV_0}{m}}$$

After prolong irradiation, potential of ring will change

$$\frac{hc}{\lambda}=\frac{hc}{\lambda_0}+eV_{ring} \qquad \Rightarrow \qquad V_{ring}=\frac{hc}{e}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)$$

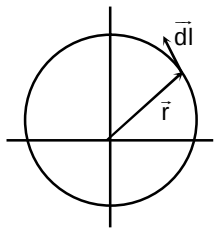
'n' : Total no. of photo electrons emitted then

$$\frac{K(ne)}{r}=\frac{hc}{e}\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)-(V_0)\Rightarrow (D)$$

6. A, B

$\vec{E}=\frac{\vec{F}}{q}\Rightarrow$ (A)

Assume Q is at origin, consider a circular loop of radius 'r', $\vec{r}$ will be perpendicular to $\vec{dl}$



As $\vec{E}$ is along $\vec{r}$

So $\int \vec{E}.\vec{dl}=0$

Consider sphere of radius 'r', $\vec{E}$ will be along $\vec{ds}$ at any position

$$\oint \vec{E}.\vec{ds}=\phi(E)(ds)=E\oint ds=\frac{Q(1-\sqrt{\alpha r})}{4\pi\in_0 r^2}(4\pi r^2)=\frac{Q}{\in_0}(1-\sqrt{\alpha r})$$

7. A, C

$$AD=R\sin\frac{\theta}{2}=r\sin\left(90°-\frac{\theta}{2}\right)$$

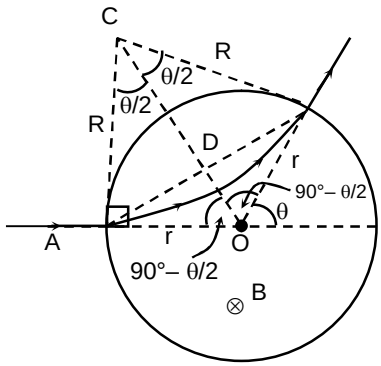
$$\Rightarrow \qquad \tan\frac{\theta}{2}=\frac{r}{R}$$

Also, $R=\frac{mv}{qB}$

$$\Rightarrow \qquad \tan\frac{\theta}{2}=\frac{rqB}{mv}$$

Angular velocity, $\omega=\frac{v}{R}=\frac{qB}{m}\rightarrow$ constant for all α-particles

So, time period will higher for the particle whose. 'θ' is higher, i.e., velocity is lower.



8. A, B, C, D

Voltage leads current (see at t = 0)

If $I = 400 \sin \omega t$ $\quad [I_0 = 400]$ (in mA)

Then $V = 200 \sin(\omega t + \phi)$ $\quad [V_0 = 200]$ (in volt.)

At t = 0, V = 100 $\Rightarrow$ $\phi=\frac{\pi}{6}$

$$Z=\frac{V_{rms}}{I_{rms}}=\frac{200/\sqrt{2}}{4\sqrt{2}}=500\Omega$$

Also, $\frac{R}{Z} = \cos\phi \quad \Rightarrow \quad R = 250\sqrt{3}\Omega$

$R^2 + (X_L - X_C)^2 = Z^2 \Rightarrow X_L = 324\ \Omega$

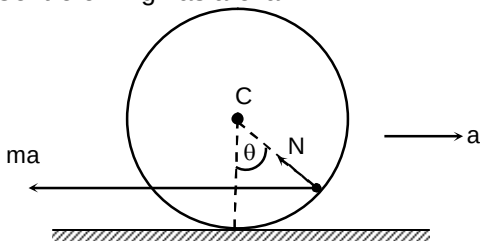
Average power = $I_{rms}\ V_{rms} \cos\phi$

$= 20\sqrt{3}W$

9. B

10. C

Sol. Centre of ring has are 'a'



a : acceleration ring w.r.t. ground

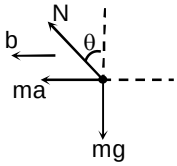
b : acceleration of particles w.r.t. ring

Hence w.r.t. 'C', for the particle

$N\cos\theta = mg$ ....(i)

$N\sin\theta + ma = mb$ ....(ii)

And for the ring



$\Rightarrow$ $N\sin\theta - f = Ma$ ....(iii)

$f.R = MR^2 .a/R$

$\Rightarrow$ $f = Ma$ ....(iv)

$\Rightarrow$ $N\sin\theta = 2Ma$

$N = mg$

$mg.\theta = 2\ Ma$

$\Rightarrow$ $a = \frac{mg}{2M}\theta$

From (ii)

$N\sin\theta + ma = mb$

$\Rightarrow$ $2Ma + ma = mb \Rightarrow a(2M + m) = mb$

$\Rightarrow$ $\frac{mg}{2M}(2M+m)\theta = m.b$

Put $\theta = \frac{x}{R}$ [x: displacement of particle w.r.t. ring]

$\Rightarrow$ $\frac{g}{2M}(2M+m)\frac{x}{R} = b \quad \Rightarrow \quad b = \frac{g}{R}\left(1+\frac{m}{2M}\right).x$

$\omega = \sqrt{\frac{g}{R}\left(1+\frac{m}{2m}\right)}$

Part – 2

$a = mb/(m + 2M)$

$\Rightarrow$ $y = \frac{mx}{m+2M}$

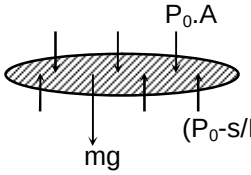
[y: displacement of ring w.r.t. ground]

$\Rightarrow$ $y = \frac{mR\theta}{m+2M}$

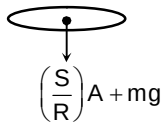
SECTION – C

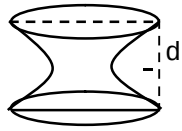
11. 9

If we draw the FBD of upper plate, we see



i.e.





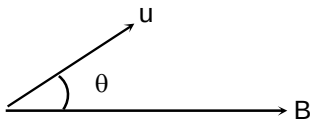
R=d/2

This force "(S/R) A + mg" must be balanced by normal reaction at same points from concern plate.

Hence N = (S/R)A + mg $\Rightarrow$ N = 90

Hence n = 9

12. 6

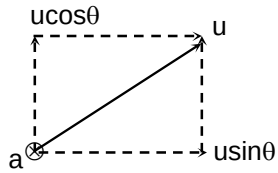


$$R = \frac{m(\sin\theta)}{qB}$$

Acceleration of particle w.r.t. an observer moving with u cos θ (along $\vec{B}$) is

$$a = \frac{(u\sin\theta)^2}{R}$$

Since this observer is not accelerated, hence net acceleration of particle is also $\frac{(u\sin\theta)^2}{R}$, which is perpendicular to its net speed 'v'.

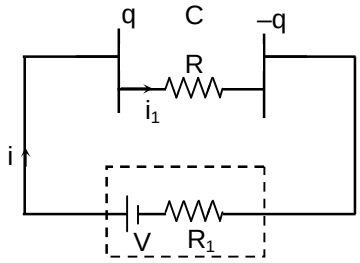


Hence ROC $= \frac{u^2}{a}$

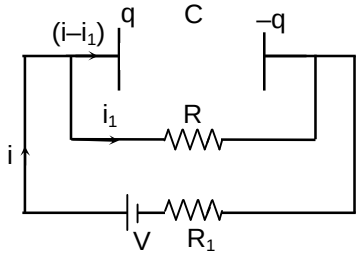
$$= \frac{R}{\sin^2\theta} = \frac{\frac{mu\sin\theta}{qB}}{\sin^2\theta} = \frac{mu}{qB\sin\theta}$$

Putting values, we get ROC as 6 m.

13. 2



$\Rightarrow$



Both the circuits are equivalent, since

$q/c = i_1R$ (i) and $i - i_1 = dq/dt$ (ii)

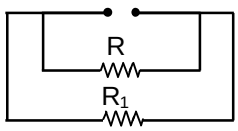
Satisfies both

Hence, time constant of circuit will be

τ = Reff. C

Where Reff = $R.R_1 / R + R_1$

Thus $\tau = \left(\frac{RR_1}{R+R_1}\right)C$



Putting $C = \frac{\in_0 \in_r A}{d}$ and $R = \frac{\rho d}{A}$

We get $\tau = 100 \in_0$

$\Rightarrow$ n = 2

14. 6

Electrical energy is present everywhere except the space occupied by conductors. When the charge is taken to infinity, energy will be present in the entire space surrounding it. Hence work done by external will be equal to the energy in the missing space.

$$W_{ext} = \frac{kq^2}{2}\left(\frac{1}{1}-\frac{1}{2}\right)+\frac{kq^2}{2}\left(\frac{1}{3}-\frac{1}{4}\right)+.....$$

$$= \frac{kq^2}{2}\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+.....\right) = \frac{kq^2}{2}\ell n2$$

$$= kq^2 \ell n\sqrt{2}$$

Hence n = 6.

15. 8

$B \times 2\pi r = \mu_0 NI$

$\Rightarrow B = \frac{\mu_0 NI}{2\pi r}$

Flux through small element = $\frac{\mu_0 NI}{2\pi r}.hdr$

$\Rightarrow$ Flux through a cross-section $= \frac{\mu_0 NIh}{2\pi}\int_a^b \frac{dr}{r}$

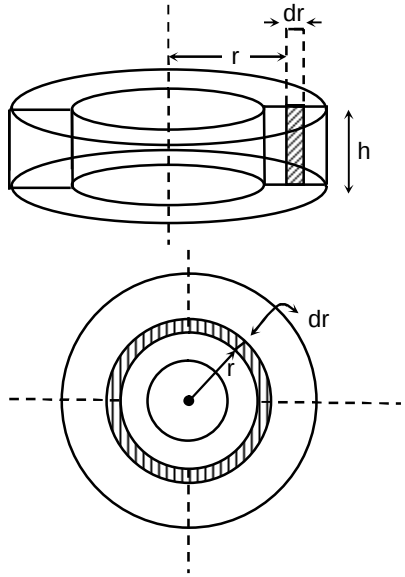
$$= \frac{\mu_0 NIh}{2\pi}\ell n\left(\frac{b}{a}\right)$$

$\Rightarrow$ Flux through 'N' turns $= \frac{\mu_0 N^2 Ih}{2\pi}\ell n\left(\frac{b}{a}\right)$

$\Rightarrow L = \frac{\phi}{I} = \frac{\mu_0 N^2 h}{2\pi}\ell n\left(\frac{b}{a}\right)$

Putting values, we get L = 0.8

Hence n = 8



16. 4

$dQ = dU + dW$

$\Rightarrow ncdT = nc_v dT + Pdv$

$\Rightarrow n.2RdT = n.\frac{3}{2}RdT + \frac{nRT}{V}dv$

$\Rightarrow \int\frac{dv}{v} = \int\frac{dT}{2T}$

$\Rightarrow \frac{v^2}{T} = constt.$

Thus, on doubling volume, temperature changes 4 times.

17. 4

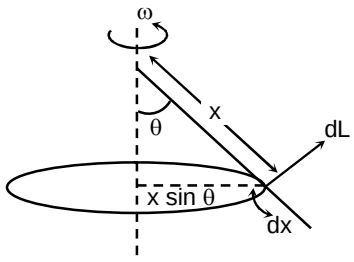
$dL = (dm)(\omega.x\sin\theta)x$

$= \frac{m}{\ell}dx.\omega x^2\sin\theta$

$\Rightarrow \quad dL = \frac{m\omega\sin\theta}{\ell}\int_0^{\ell} x^2dx$

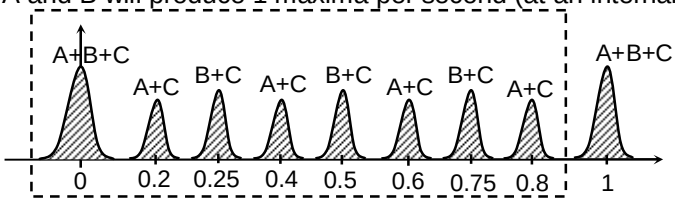
$\Rightarrow \quad L = \frac{m\ell^2\omega\sin\theta}{3}$

Putting values, we get L = 4.



18. 8

A and C will produce 5 maxima per second (at an internal of 0.2 sec)
B and C will produce 4 maxima per second (at an internal 0.25 sec)
A and B will produce 1 maxima per second (at an internal 1 sec)



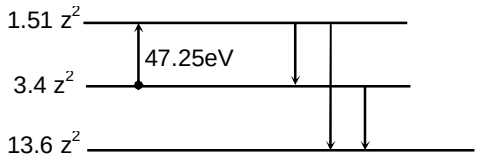
Total no. of maxima in 1 sec is 7 secondary + 1 primary = 8 maxima

19. 5

As we are getting 3 different photons so transition must be from n = 3 to n = 1 and energies of these photons are more than or equal to 47.25 so initially electrons must be in n = 2

$3.4\ z^2 - 1.51z^2 = 47.25$

$\Rightarrow \quad z = 5$



20. 5

$\frac{n_2}{V} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

$\frac{n_2}{V} = \frac{n_2 - n_1}{R} + \frac{n_1}{u}$

$u = -x \quad \Rightarrow \quad \frac{n_2}{v} = \frac{n_2 - n_1}{R} - \frac{n_1}{x}$

For real image V must be +ve

$\frac{n_2 - n_1}{R} - \frac{n_1}{x} > 0$

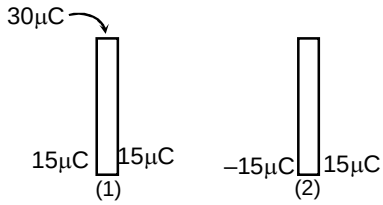
$\Rightarrow \quad x > \frac{n_1R}{n_2 - n_1}$

SECTION – D

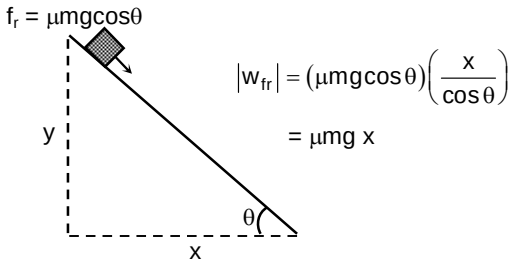
21. 01800.00

Charge is initially distributed as shown in figure. Now after connecting the battery, potential difference across the plates of the capacitor should also become 40 V in steady state.

Hence, $Q_f = CV$
$= (1.5)(40)\ \mu C$
$= 60\ \mu C$ (Charge on inner surface)
Hence, charge supplied by battery is $45\mu C$
So, work done by battery, $W = (Q)(V)$
$= (45\mu C)(40V)$
$= 1800\ \mu J$
So, answer is 1800.



22. 00720.00
Hence, work done by friction is independent of angle of inclination. It only depends upon horizontal displacement.
So, $(W_{fr}) = (0.2)(18)(10)(20)$
$= 720$ J



23. 00004.00
Given that $C = C_V + 4R$ .....(1)
Compare with $dQ = dU + PdV$
$CdT = C_V d_T + Pd_V$

$$C = C_V + P\frac{dV}{dT} \quad .....(2)$$

We get, $4R = \frac{PdV}{dT}$

$$4R = \frac{RT}{V}\frac{dv}{dT}$$

$$4\int\frac{dT}{T} = \int\frac{dv}{v}$$

$$\ell nT^4 = \ell nV$$

$$VT^{-4} = \text{const}$$

Hence, n = 4

Chemistry

PART – II

SECTION – A

24. B

$P_A = P_B$

$XA \cdot P_A^0 = X_B \cdot P_B^0$

$$\frac{X_A}{X_B} = \frac{P_B^0}{P_A^0}$$

$$1 + \frac{X_A}{X_B} = 1 + \frac{P_B^0}{P_A^0}$$

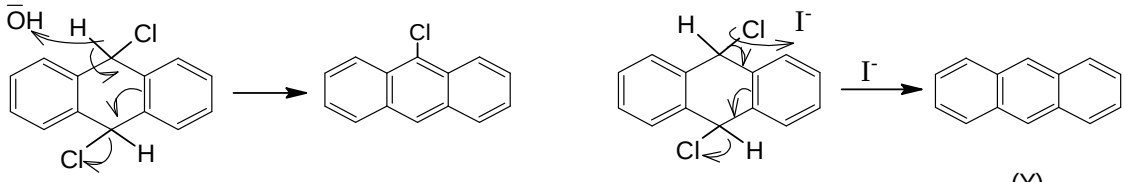
$$\frac{1}{X_B} = \frac{P_A^0 + P_B^0}{P_A^0}$$

$$X_B = \frac{P_A^0}{P_A^0 + P_B^0}$$

Similarly, $X_A = \frac{P_B^0}{P_A^0 + P_B^0}$

$$P_T = P_A^0 . X_A + P_B^0 . X_B = \frac{2P_A^0 . P_B^0}{P_A^0 . P_B^0}$$

25. D



26. D

27. AC

Since 4 milli mole of NaOH was required to reach first equivalence point, the molecular wt. was

$$\frac{664\,mg}{4\,m\,mol} = 166$$

60 ml of NaOH is represented by a point midway to second equivalence point, at which

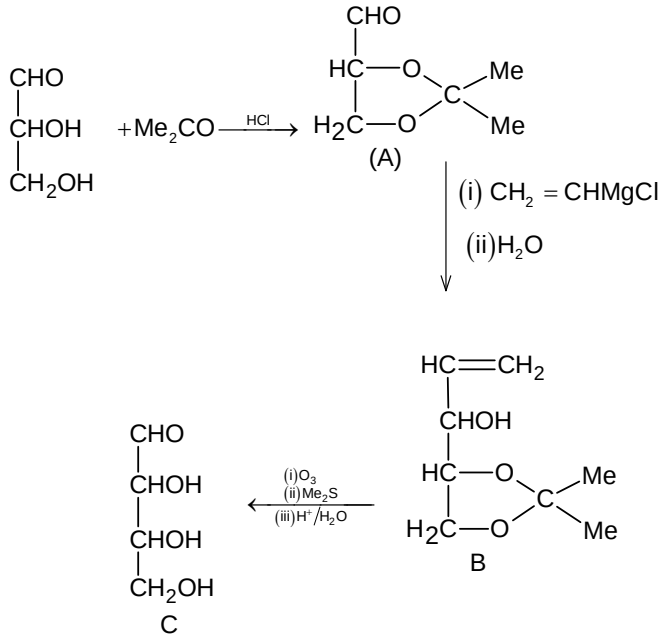
$pH = Pka_2 = 8.08$

At the first equivalence point,

$$pH = 5.85 = \frac{Pka_1 + Pka_2}{2}$$

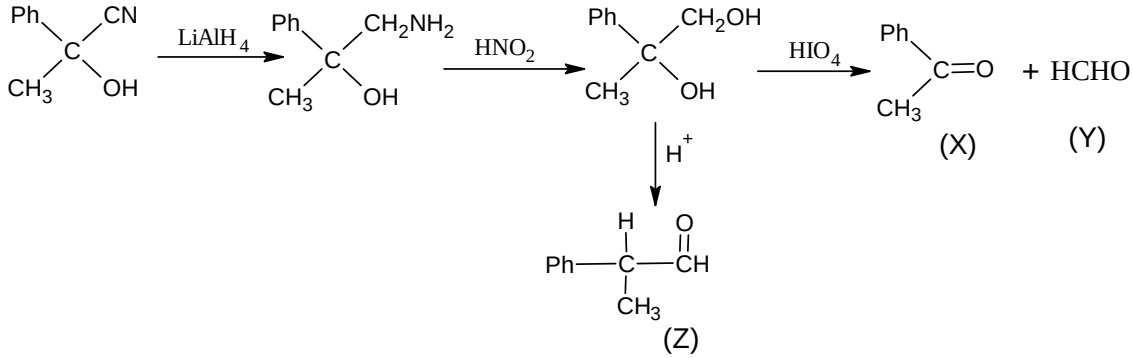
$\therefore\ Pka_1 = 3.62$

28. A, B, C, D



29. B C

30. AC



31. BC
32. C
33. B

Sol. (for 32–33)

$CuSO_4.5H_2O\ (s) \xrightarrow{\Delta} CuSO_4(s)$

(A) Blue (B) white

$2CuSO_4 + 4KI \longrightarrow \underset{\text{ppt}}{Cu_2I_2\downarrow} + I_2\uparrow + 2K_2SO_4$

$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

millimoles of thiosulphate required = $0.25 \times 20 = 5$

$\therefore$ millimoles of I_2 used = 5/2

$\therefore$ millimoles of $CuSO_4$ dissolved = $\frac{5}{2} \times 2 = 5$

$\therefore$ $(wt)_{CuSO_4}$ = $5 \times 10^{-3} \times 159.5 = 0.797$ gm in 100 mL of solution.

Hence in 1 L solution (wt.) of $CuSO_4$ dissolved = 7.97 gm

SECTION – C

34. 3

$$n_{M^{n+}} = \frac{1.92}{2 \times 60} mol = 0.016\, mol$$

$$nhydrazinehydrate = \frac{0.9 \times 5.332}{50} mol = 0.096 mol$$

$n_{eq.}$ KIO_3 = $4 \times \frac{1}{10} N \times 0.48L$ = 0.192 eq. = $n_{eq.}$ hydrazine reacted with KIO_3

$\therefore$ n hydrazine left after reaction with M^{n+} = $\frac{0.192}{4} mol = 0.048 mol$

$\therefore$ n hydrazine hydrate reacted with M^{n+} = (0.096 – 0.048) mol = 0.048 mol

M^{n+} + n $H_2NNHCOO^-$ $\rightarrow$ [M $(H_2NNHCOO)_n$]

$$\frac{n_{M^{n+}}}{n_{hydrazine}} = \frac{1}{n} = \frac{0.016}{0.048} = \frac{1}{3}$$

$\therefore$ n = 3

35. 3

Cu^{+2}, Cd^{+2}, Ag^+

36. 4

$$\frac{y_A}{y_B} = \frac{P_A^\circ}{P_B^\circ}$$

$$\frac{n_A}{n_B} = \frac{P_A^\circ}{P_B^\circ}$$

$$\frac{\frac{x}{378}}{\frac{1}{18}} = \frac{120}{630}$$

Hence $x = \frac{378 \times 4}{21 \times 18} = 4$

37. 2

Mole fraction of solute (x) = $\frac{\left(\frac{66.9}{148.3}\right)}{\left(\frac{66.9}{148.3}\right) + \left(\frac{100}{18}\right)} = 8.03 \times 10^{-3}$

Relative lowering of vapour pressure

$$\frac{P^0 - P_s}{P^0} = (x \times i) \;\Rightarrow\; \frac{\Delta P}{P^0} = x \times i$$

$$= \frac{760 - 747}{760} = 8.03 \times 10^{-3} \times i$$

$M(NO_3)_n$ $\rightarrow$ M^{n+} + $n\,NO_3^-$

1-α α $n\alpha$

Van't Hoff Factor $i = (1 + n\alpha) = [1 + 0.565n]$

$i = \dfrac{13}{760 \times 8.03 \times 10^{-3}} \simeq 2.13$

$\therefore$ $1 + 0.565\, n = 2.13$

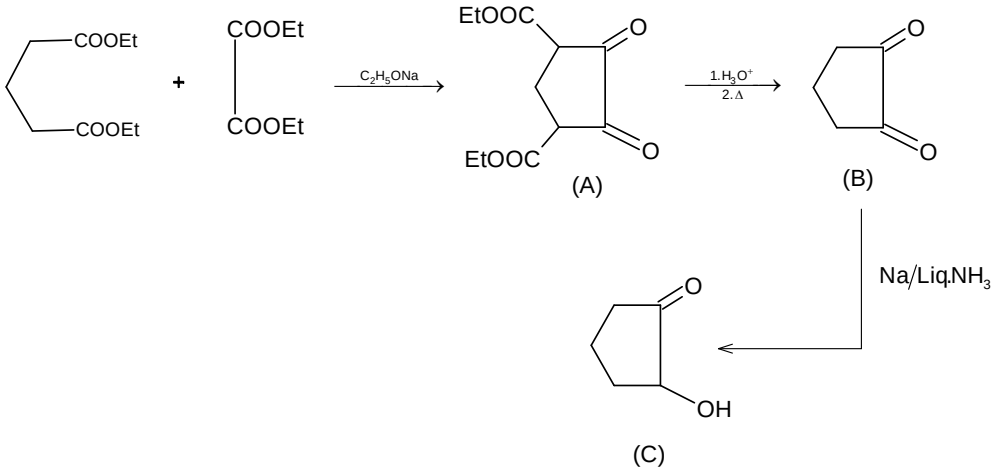
$\therefore$ $n = 2$

38. 6

39. 3

$H_3PO_2, H_4P_2O_5, H_3PO_3$

40. 2



41. 6

$\dfrac{1}{3}\dfrac{d(H_2)}{dt} = k[NH_3]^0$

$\dfrac{d(H_2)}{dt} = 3 \times 2 = 6 M min^{-1}$

42. 4

$XeF_2, XeOF_4, BrCl_3, \left[Fe(CN)_6\right]^{-4}$

43. 4

(i) all given sulphides are less soluble than MnS, hence x = 0

(ii) $Pb(OH)_2$, $Zn(OH)_2$, $Cu(OH)_2$, $Fe(OH)_2$, are less soluble than $Mg(OH)_2$, y = 4

Hence ; x + y = 0+4 = 4

SECTION – D

44. 00004.00

Anode half cell $H_2 \rightarrow 2H^+ + 2e^-$

cathode half cell $Hg_2Cl_{2\,(s)} + 2e^- \rightarrow Hg(l) + 2Cl^-$

Net cell reaction $Hg_2Cl_{2(s)} + H_2 \rightarrow 2H^+ + Hg(l) + 2Cl^-$

$\therefore$ $E_{cell} = E^0_{cell} - \dfrac{0.0591}{2}\log\dfrac{\left[H^+\right]^2 \times \left[Cl^-\right]^2}{P_{H_2}}$

$0.5164 = 0.28 - \dfrac{0.0591}{2}\log\left[H^+\right]^2$; $P_{H_2} = 1 atm$, $Cl^- = 1M$

$0.2364 = -0.0591 \log[H^+]$

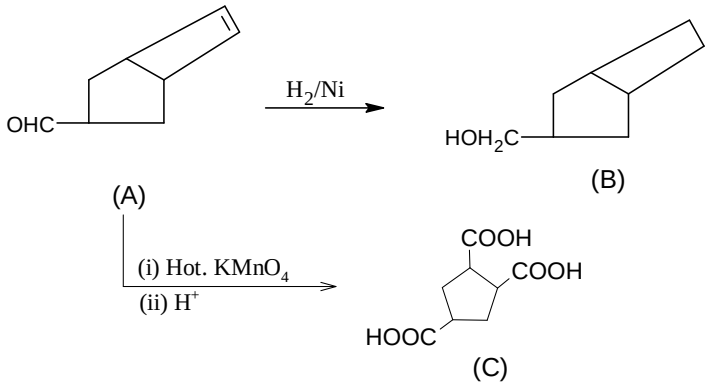
$$pH = \frac{0.2364}{0.0591} \simeq 4 = \frac{pk_1 + pk_2}{2}$$

In order to maintain the pH of left hand half cell the vol. of NaOH required

$\therefore$ $0.1 \times 100 = 2.5 \times V$ (1st equivalence point)

$$V = \frac{10}{2.5} = 4mL$$

45. 00004.50



$\therefore$ A is $C_8H_{10}O$

$\therefore$ x = 8, y = 10 and z = 1

46. 00001.25

For an adiabatic process,

$TV^{\gamma-1}$ = constant

$$\therefore \quad T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} = \frac{T_1}{2}(5.66V_1)^{\gamma-1}$$

Hence, $2 = (5.66)^{\gamma - 1}$

or $\log 2 = (\gamma - 1) \log 5.66$

$\therefore$ $\gamma = 1.4$

The gas is, therefore a diatomic gas and have five degrees of freedom.

The work done by a gas during an adiabatic process is

$$W = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

Since $P_1V_1^{\gamma} = P_2V_2^{\gamma} = P_2(5.66V_1)^{\gamma}$

$$P_2 = \frac{P_1}{(5.66)^{\gamma}}$$

$$\therefore \quad W = \left[\frac{P_1(5.66V_1)}{(5.66)^{\gamma}} - P_1V_1\right]\frac{1}{0.4} = 1.25\ P_1V_1$$

or $W = 1.25\ PV$

Hence; x = 1.25

Mathematics

PART – III

SECTION – A

47. B

Sol. f(x) = f(4-x), f(x) = f(x+10), f(x) = f(14-x). There are 201 multiple of 10 and 200 numbers that remainders have 4 divided by 10.

48. B

Sol. $a_{n+2} = 1 - \frac{1}{a_{n+1}} = 1 - \frac{a_n}{a_n - 1} = \frac{1}{1 - a_n}$

$a_{n+3} = 1 - \frac{1}{a_{n+2}} = 1 + (a_n - 1) = a_n$

So, $\{a_n\}$ is periodic sequence with period 3

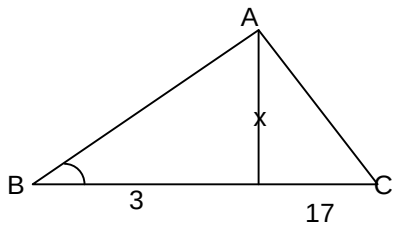
$a_1 = 2 \; a_2 = \frac{1}{2} \; a_3 = -1$

$p_3 = -1$

$P_{2009} = (P_3)^{669} . P_2 = (-1)^{669} = -1$

49. A

Sol. $\tan^{-1}\left(\frac{22}{7}\right) = \tan^{-1}\frac{3}{x} + \tan^{-1}\frac{17}{x}$



$\frac{22}{7} = \frac{\frac{3}{x} + \frac{17}{x}}{1 - \left(\frac{3}{x}\right).\left(\frac{17}{x}\right)}$

$11x^2 - 70x - 56 = (x - 11)(11x + 51) = 0$

$\Rightarrow x = 11$

$\Delta = \frac{1}{2}(3 + 17)(11) = 110$

50. B, D

Sol. We have $f(x) = x^2 + ax^2 + bx^3$

Where $a = \int_{-1}^{1} t.f(t)dt$ and $b = \int_{-1}^{1} f(t)\,dt$

Now, $a = \int_{-1}^{1} t\left[(a+1)t^2 + bt^3\right]dt$

$a = 2b\int_{0}^{1} t^4 dt = \frac{2b}{5}$ ...(1)

Again $b = \int_{-1}^{1} f(t)\,dt = \int_{-1}^{1}\left[(a+1)t^2 + bt^3\right]dt$

$= 2\int_{0}^{1}(a+1)t^2\,dt$

$b = \frac{2(a+1)}{3}$ ...(ii)

From equations (1) and (ii)

$\frac{5a}{2} = \frac{2(a+1)}{3}$

$\left(\frac{5}{2} - \frac{2}{3}\right)a = \frac{2}{3}$

$\frac{11}{6}a = \frac{2}{3}$

$a = \frac{4}{11}$ and $b = \frac{10}{11}$

Hence $\int_{-1}^{1} t.f(t)\,dt = \frac{4}{11}$ and $\int_{-1}^{1} f(t)\,dt = \frac{10}{11}$

$\therefore f(x) = (a+1)x^2 + bx^3$

$\left.\begin{aligned} f(1) &= (a+1)+b \\ f(-1) &= (a+1)-b \end{aligned}\right]$

$f(1) + f(-1) = 2(a+1) = \frac{30}{11}$

And $f(1) - f(-1) - 2b = \frac{20}{11}$

51. A, C

Sol. $A^3 + B^3 + C^3 = A^3 + B^3 + C^3 - 3ABC$

$= (A + B + C)(A^2 + B^2 + C^2 - AB - BC - CA)$

$= (A + B + C)(A + \omega B + \omega^2 C)(A + \omega^2 B + \omega C)$ {ω = cube root of unity}

$= (A + B + C)(A + \omega B + \omega^2 C)\left(\overline{A + \omega B + \omega^2 C}\right)$

then $\det(A^3 + B^3 + C^3) \cdot \det(A + B + C) = \det(A + B + C)^2 \cdot \det(A + \omega B + \omega^2 C)\,\overline{\det\left(A + \omega B + \omega^2 C\right)}$

$= (\det(A + B + C))^2 \cdot |\det(A + \omega B + \omega^2 C)|^2 \geq 0.$

52. A, D

Sol. The given variable line can be expressed as

$\Rightarrow 3x + 4y - 7 + \lambda(x - 2y + 1) = 0$

$\Rightarrow L_1 + \lambda L_2 = 0$

$L_1 \equiv 3x + 4y - 7 = 0, L_2 \equiv x - 2y + 1 = 0$

Point of intersection of $L_1 = 0$ and $L_2 = 0$ is (1, 1)

$\therefore$ a = b = 1 $\Rightarrow a + 2b = 3$

Now, $l = \lim_{x \to 0^-} \frac{[\sin x] - 2 + \{\cos x\}}{\{x\} - 1}$

$= \lim_{h \to 0} \frac{[\sin(0-h)] - 2 + \{\cos(0-h)\}}{\{0-h\} - 1}$

$= \frac{-1-2+1}{1-1} = \frac{-2}{0}$

Hence l does not exist

53. A, B

Sol. $f(3n) = f(f(f(n))) = 3f(n), \forall\ n \in N$

put n = 1, f(3) = 3f(1)

$f(1) \neq 1$ as if f(1) = 1, then

f(f(1)) = 3 . 1 = 3 = f(1) = 1 which is incorrect, thus f(1) > 1 and f(f(1)) > f(1)
3 = f(f(1)) > f(1) > 1
so f(1) = 2
f(2) = f(f(1)) = 3.

54. B, C, D

Sol. As, $\vec{r}_3 = (\vec{r}_1 \times \vec{r}_2) \times \vec{r}_2 = (\vec{r}_1.\vec{r}_2)\vec{r}_2 - (\vec{r}_2.\vec{r}_2)\vec{r}_1$

$= 6(2\hat{i} + \hat{j} + \hat{k}) - 6(\hat{i} + \hat{j} + 3\hat{k}) = (6\hat{i} - 12\hat{k})$

$\therefore$ Row 3 vector $\pm\frac{6(\hat{i} - 2\hat{k})}{6\sqrt{5}}.\sqrt{5} = \hat{i} - 2\hat{k}$ or $-\hat{i} + 2\hat{k}$

$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$ or $\begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} = \text{Tr.}(A) = 0, 4.$

Also, $[\vec{r}_2\ \vec{r}_3\ \vec{r}_2 \times \vec{r}_3] = |\vec{r}_2||\vec{r}_3||\vec{r}_2 \times \vec{r}_3|$

$= \sqrt{6}.\sqrt{5}.\sqrt{6}.\sqrt{5} = 30$

Since $\vec{r}_1, \vec{r}_2$ and $\vec{r}_3$ are coplanar

$\Rightarrow$ they are linearly dependent

$\therefore \left[\vec{r}_1 \times \vec{r}_2\ \vec{r}_2 \times \vec{r}_3\ \vec{r}_3 \times \vec{r}_1\right] = \left[\vec{r}_1\ \vec{r}_2\ \vec{r}_3\right]^2 = 0$

55. C

Let $A \equiv (-a, 0), B \equiv (a, 0)$, the equations of the circles are $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ .....(1)

Any tangent to (1) is $y = mx + b\sqrt{1 + m^2}$

Let (h, k) be the coordinates of the focus, A and B lie on the parabola.

$\therefore \sqrt{1 + m^2}\sqrt{(h + a)^2 + k^2} = -ma + b\sqrt{1 + m^2}$

And $\sqrt{1 + m^2}\sqrt{(h - a)^2 + k^2} = ma + b\sqrt{1 + m^2}$

Eliminating m from the above equations, we get the locus as $\frac{x^2}{b^2} + \frac{y^2}{b^2 - a^2} = 1$

56. A

For $a = 4, b = 3$, the curve becomes $\frac{x^2}{9} - \frac{y^2}{7} = 1$

The length of LR = $\frac{2B^2}{A} = \frac{2 \times 7}{3} = \frac{14}{3}$.

SECTION – C

57. 4

Sol. $L : 3^4 - [{}^3C_1(2^4 - 2) + {}^3C_2] = 36$

M : If $x > 0, \text{sgn}(x) = 1$

$f(x) = 0 + \frac{\pi}{2} = \frac{\pi}{2}$

For $x = 0$ $f(x)$ is not defined

$\therefore$ For $x < 0, f(x) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$

$\therefore$ M = 1

N : Coefficient of t^5 = coefficient of t^2 in $(1 + t^2)^5 \times$ coefficient of t^3 in $(1 + t^3)^8$

$= 5 \times 8 = 40$

$\Rightarrow LM+N=36+40=76$

58. 8

Sol. $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)=\prod_{n=3}^{\infty}\left(\frac{n^2-4}{n^2}\right)$

$=\prod_{n=3}^{\infty}\left(\frac{n-2}{n}\right)\times\prod_{n=3}^{\infty}\left(\frac{n+2}{n}\right)$

$=\frac{n\left(1+\frac{1}{n}\right)n\left(1+\frac{2}{n}\right)}{n^2\left(1-\frac{1}{n}\right)}\times\frac{2}{3.4}=\frac{1}{6}$

$\Rightarrow L=\frac{1}{6}$

$M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)=M=\prod_{n=2}^{\infty}\frac{n-1}{n+1}.\frac{(n^2+n+1)}{(n^2-n+1)}$

$=\lim_{x\to\infty}\frac{2}{n(n+1)}.\frac{n^2+n+1}{3}=\frac{2}{3}$

$\Rightarrow M=\frac{2}{3}$

$N=\prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}=\prod_{n=1}^{\infty}\left(\frac{n+1}{n}\right)^2\left(\frac{n}{n+2}\right)$

$=\frac{2(n+1)}{n+2}=2$

$\Rightarrow N=2$

$L^{-1}+M^{-1}+N^{-1}=6+\frac{3}{2}+\frac{1}{2}=8$

59. 2

Sol. Let $S=\sum_{n=1}^{99}\sqrt{10+\sqrt{n}}$ $\qquad T=\sum_{n=2}^{99}\sqrt{10-\sqrt{n}}$

$\sqrt{2}S=\sum_{n=1}^{99}\sqrt{20+2\sqrt{n}}=\sum_{n=1}^{99}\left(\sqrt{10+\sqrt{100-n}}+\sqrt{10-\sqrt{100-n}}\right)$

$\sqrt{2}S=\sum_{n=1}^{99}\left(\sqrt{10+\sqrt{n}}+\sqrt{10-\sqrt{n}}\right)=S+T$

$\sqrt{2}S-S=T$

$\frac{S}{T}=\sqrt{2}+1$

60. 5

Sol. For each 3-element subsets x = {a, b, c} of the set M = {1, 2, ...23} with a + b + c < 36, map f(x) = {24 – a, 24 – b, 24 – c} then f is a bijection the set C_1 of all three element subsets of M with a + b + c < 36 and set C_2 of all three element subsets M with a + b + c > 36, Hence $|C_1|=|C_2|$.

Total $={}^{23}C_3=1771$ different 3 element subset of M, $|C_1|=\frac{1}{2}(1771-61)=855$

61. 6

Sol. Given, $f(x^2).f''(x)=f'(x).f'(x^2)$

Put $x=1$ in the given relation

$f(1).f''(1)=(f'(1))^2$

Let $f'(1) = a$ and $f''(1) = b$

$\therefore \quad b = a^2 \quad ....(1)$

Differentiating the given relation

$f(x^2).f'''(x) + f''(x).2x f'(x^2)$

$= f''(x) f'(x^2) + f'(x).(2x) f''(x^2)$

Put $x = 1$

$8 + 2ba = 2ab + ab$

$\Rightarrow \quad ab = 8 \quad ....(2)$

62. 3

Sol. Since roots of $x^2 - x - 1 = 0$, are $P = \frac{1}{2}(1+\sqrt{5})$ and $q = \frac{1}{2}(1-\sqrt{5})$

$ax^{17} + bx^{16} + 1 = 0$

$ap^{17} + bp^{16} = -1 \quad ...(1)$

$aq^{17} + bq^{16} = -1 \quad ...(2)$

Multiply (1) by q^{16} and (2) by p^{16}

$ap + b = -q^{16}$

$aq + b = -p^{16}$

$a = \frac{p^{16} - q^{16}}{p - q} = (p^8 + q^8)(p^4 + q^4)(p^2 + q^2)(p + q)$

$p + q = 1 \qquad p^4 + q^4 = (p^2 + q^2)^2 - 2(pq)^2 = 9 - 2 = 7$

$p^2 + q^2 = 3 \qquad p^8 + q^8 = (p^4 + q^4)^2 - 2(pq)^4 = 49 - 2 = 47$

$a = 47 \times 7 \times 3 \times 1 = 987$

63. 5

Sol. $J = \int_0^1 (1 - x^{50})^{100} x^{100} \, dx$

$= \left((1 - x^{50})^{100} \frac{x^{101}}{101} \right)_0^1 - \int_0^1 100(1 - x^{50})^{99} (-50x^{49}) \frac{x^{101}}{101} dx$

$= 0 + \frac{5000}{101} \int_0^1 (1 - x^{50})^{99} (1 - (1 - x^{50})) x^{100} \, dx$

$= \frac{5000}{101} \left(\int_0^1 (1 - x^{50})^{99} x^{100} dx - \int_0^1 (1 - x^{50})^{100} x^{100} \, dx \right)$

$\frac{5101}{101} J = \frac{5000}{101} I$

$\Rightarrow \frac{I}{J} = \frac{5101}{5000} = \frac{m}{n}$

$\therefore \quad \frac{m - 1 - n}{20} = \frac{5101 - 1 - 5000}{20} = 5$

64. 1

Sol. $D_1 = |10| \qquad D_2 = \begin{vmatrix} 10 & 3 \\ 3 & 10 \end{vmatrix} = 10 \times 10 - 3.3 = 91$

$$D_3 = \begin{vmatrix} 10 & 3 & 0 \\ 3 & 10 & 3 \\ 0 & 3 & 10 \end{vmatrix} = 10\begin{vmatrix} 10 & 3 \\ 3 & 10 \end{vmatrix} - 3\begin{vmatrix} 3 & 3 \\ 0 & 10 \end{vmatrix} + 0\begin{vmatrix} 3 & 10 \\ 0 & 3 \end{vmatrix}$$

$D_n = 10D_{n-1} - 9D_{n-2} = 10(D_{n-1} - D_{n-2}) + D_{n-2}$

$D_2 = 10D_1 - 9D_0 \Rightarrow D_0 = 1$ by recursion formula $D_4 = 7381$

$D_1 - D_0 = 10 - 1 = 9$

$D_2 - D_1 = 81 = 9^2$

$D_3 - D_2 = 820 - 91 = 729 = 9^3$

$D_4 - D_3 = 7381 - 820 = 6561 = 9^4$

$$D_n = D_0 + 9^1 + 9^2 + ...9^n = \sum_{i=0}^{n} 9^i$$

Desired sum

$$\sum_{n=1}^{\infty} \frac{1}{\frac{8.9^{n+1}-1}{8}+1} = \sum_{n=1}^{\infty} \frac{1}{9^{n+1}-1+1} = \sum_{n=1}^{\infty} \frac{1}{9^{n+1}} = \frac{1}{72}$$

p + q = 1 + 72 = 073

65. 2

Sol. For n = 0, 1, 2..... Let a_n be the prob. That bug is at vertex A after crawling exactly n meter's, then

$$a_{n+1} = \frac{1}{3}(1 - a_n)$$

Become the bug can be at vertex A after crawling n + 1 meter if and only if
(i) It was not at A following crawling of n meter (this has prob $1 - a_n$)
(ii) from one of other vertices it heads towards A (thes probability 1/3)

Since $a_0 = 1$ $\quad a_1 = 0$ $\quad a_2 = \frac{1}{3}$ $\quad a_3 = \frac{2}{9}$

$a_4 = \frac{7}{27}$ $\quad a_5 = \frac{20}{81}$ $\quad a_6 = \frac{61}{243}$ $\quad a_7 = \frac{182}{729}$

66. 4

Multiple the given equation by $z^3 - 1 = 0$

Yield $z^9 - 1 = 0$

$z_n = \cos n, 40^o + i \sin n\ 40^o n = 0, 1, 2....., 8$ of these only z_3 and z_4 are in the second quadrant. However, Since the solution of $z^9 - 1 = 0$ are distinct and since z_3 is a solution of $z^3 = 1$ It cannot be solution of the original equation. It follows that desired root in z_4 with of agree measure 160°.

SECTION – D

67. 18564.00

Sol. The sum as $\left(\binom{6}{0} + \binom{6}{1}x + \binom{6}{2}x^2 + \binom{6}{3}x^3 + ... + \binom{6}{6}x^6\right)^3$.

We can see the sum we wish to compute is just the coefficient of the x^6 term. However

$P(x) = ((1+x)^6)^3 = (1+x)^{18}$. Therefore, the coefficient of the x^6 term is just $\binom{18}{6} = 18564$

68. 00038.00

Sol. The probability that the chord doesn't intersect the triangle is $\frac{11}{25}$.

The only way this can happen is if the two points are chosen on the same arc between two of the triangle vertices. The probability that a point is chosen on one of the arcs opposite one of the

base angles is $\frac{x}{180}$, and the probability that a point is chosen on the arc between the two-page angles is $\frac{180-2x}{180}$. Therefore, we can write

$$2\left(\frac{x}{180}\right)^2+\left(\frac{180-2x}{180}\right)^2=\frac{11}{25}$$

This simplifies to

$$x^2-120x+3024=0$$

Which factors as

$$(x-84)((x-36)=0$$

So $x=84,36$. The difference between.

69. 00501.00

The equation $\log(kx)=2\log(x+2)$ is valid for $kx>0$ and x> -2.

It is easy to seen that the all negative values of k make the equation $\log(kx)=2\log(x+2)$ have 500 integer values and for positive reason of x, k has only one value.